Nahush Shrivatsa

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Education

New York University — September 2025 – May 2027
Master of Science in Computer Science — *Cumulative GPA: 3.7*

University of Washington, Seattle Campus — September 2021 – June 2025
Bachelor of Science in Computer Science — *Cumulative GPA: 3.8 (Annual Dean's List)*

Technical Skills

Languages: Python, Java, C/C++, TypeScript, OCaml, SQL, Bash, YAML, HTML/CSS
Frameworks & Databases: PyTorch, Hydra, Flask, REST APIs, React Native, PostgreSQL, PostGIS, MongoDB
Infrastructure & DevOps: Linux (Ubuntu), Docker, Kubernetes, OpenShift, GitLab CI/CD, SLURM, Git, Grafana

Relevant Experience

Resilient Urban Networks Lab @ NYU — New York, NY
Graduate Researcher — *June 2026 – Present*

- Built the **FastAPI** backend and **React** visualization for an open-source agent-based simulation of recommender systems shaping urban travel, streaming live agent movement over an OpenStreetMap road network.
- First-author paper under review at **ACM GeoSim 2026**; now opening a new research direction on virality, designing and implementing the model's social-influence layer.
- Proposed, configured, and currently maintaining the lab's compute server (**Ubuntu**, **PostgreSQL** + **PostGIS**), extending the model onto large-scale census and mobility data and documented for lab-wide experiments.

NYU IT High Speed Research Network — New York, NY
Research Engineer — *January 2026 – Present*

- Characterized per-port power draw across enterprise switches and transceivers using custom-flashed energy monitors streamed to **Prometheus**; built a tool estimating total draw for any switch configuration to guide capacity planning.
- Assembled a **CaImAn + FIOLA Python** pipeline for the Fenton Lab converting neuronal fluorescence imaging into real-time activity signals, generating a temporal dataset & enabling closed-loop experimentation.

Nokia — Dubai, UAE
Software Engineer Intern II — *June 2024 – September 2024*

- Developed a Customer Experience Index distilling large KPI datasets into a single actionable score via **MongoDB** and a **Python** Flask app connected to **Grafana** dashboards, cutting ∼60% of analyst time per report.
- Led system administration of **OpenShift** platform; managed users, maintained technical documentation, and deployed 10+ containerized workloads and 4 **GitLab** CI/CD pipelines.

Software Engineer Intern I — *July 2023 – October 2023*

- Implemented a scalable distributed file store with **GPG encryption** using **Apache NiFi** and **Bash** scripting, raising transfer throughput ∼1.6× over the previous workflow.

Additional Experience

Washington Tunneling — Seattle, WA
Lead Software Engineer | GitHub — *January 2024 – June 2025*

- Sole embedded engineer for a tunnel boring machine, writing the full **C++** stack on **ESP32** microcontrollers for closed-loop real-time motor control and telemetry across the machine's sensor suite.
- Designed the serial protocol and **Python** server bridging the machine's microcontrollers to the **TypeScript** frontend, delivering low-latency command execution and real-time data feeds.

Systems Neuroscience & AI Lab (Golub Lab) — Seattle, WA
Undergraduate Researcher — *September 2024 – June 2025*

- Replicated Mante et al. (2013) with Dr. Matt Golub and Dr. Jacob Sacks, training **recurrent neural networks** to recover the low-dimensional dynamics underlying decision-making in the prefrontal cortex.
- Built a modular SLURM-friendly **PyTorch + Hydra** platform fusing custom RNNs with biomechanical simulations.

Projects

Home Server | *Docker, WireGuard, Caddy, Bash* | GitHub — October 2025

- Self-hosted 15+ **Docker** containers across isolated network stacks (Jellyfin, Seafile, Vaultwarden) on a Hyper-V VM.
- Authored Daemon & Bash automation scripts for 0-downtime rolling updates, image cleanup & instant Tailscale link.

Distributed Systems Lab | *Java, Paxos, Sharding, RPC* | GitHub — May 2023

- Developed a sharded, linearizable, highly available key-value store with dynamic load balancing using Paxos in Java.